DHEERAJ KARKI

MSc - Robotics, University of Bristol | C++ · CAD · Robotics

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PROFESSIONAL SUMMARY

Software engineer with 10+ years of experience building high-performance C++ applications in the CAD industry, specialising in computational geometry, spatial data structures, memory-efficient algorithms, system performance, and cross-platform interactive applications. Currently pursuing an MSc in Robotics at the University of Bristol to apply this strong foundation to autonomous and intelligent systems.

TECHNICAL SKILLS

Languages: C++, C, JavaScript, Python

Domains: Computational Geometry, Spatial Data Structures, CAD Systems, Geometry Processing, Rendering & Graphics, Algorithms

Frameworks & Libraries: React.js, ODA (Open Design Alliance)

Concepts: Planar Graph Traversal, Closed-Boundary Detection, Cross-Platform Development, System-Level Optimisation, Real-Time Interactive Applications

Practices: Software Architecture, Code Refactoring, Performance Optimisation, Developer Mentoring

PROFESSIONAL EXPERIENCE

Arden Software Hyderabad, India — Jan 2019 – Sep 2026

Staff Software Developer — Sep 2025 – Sep 2026

- Modernised a legacy desktop CAD application by driving migration to a modern engine, achieving feature parity across 3 platforms: web, desktop, and cloud.
- Engineered quadtree-based spatial data structures to optimise polygon lookup, point-in-polygon detection, and nested polygon-hierarchy resolution, accelerating geometric queries by approximately 35%.
- Collaborated across 3 functions—CAD, architecture, and UX—while refactoring 5+ core modules to improve maintainability and code quality.

Senior Software Developer — Oct 2021 – Sep 2025

- Architected a reusable annotation framework for the CAD drawing engine, supporting dimensions, text, callouts, and leaders to establish a scalable foundation for future drawing capabilities.
- Implemented import/export functionality for four major CAD formats (DXF, DWG, IPD, and internal JSON), expanding interoperability and seamless data exchange across tools.
- Streamlined rendering pipelines and geometry-processing algorithms to enhance platform responsiveness, usability, and geometric computation efficiency.

Software Developer — Jan 2019 – Oct 2021

- Optimised HTML5 web canvas rendering for complex CAD drawings by intercepting render calls, replaying cached geometry, and batching draw operations, reducing rendering latency by approximately 90%.

Graebert GmbH Noida, India — Feb 2016 – Jan 2019

Senior Software Engineer — Jan 2018 – Jan 2019

- Designed extensible plugin frameworks and drawing-property systems to facilitate modular and scalable CAD feature development.

Software Engineer — Feb 2016 – Jan 2018

- Developed custom entities, view systems, and interactive editing tools for a cloud-based CAD editor, improving rendering performance and user interaction workflows.

Sevya Multimedia Technology Hyderabad, India — Feb 2014 – Jan 2016

Software Developer

- Engineered C/C++ automation tools for enterprise clients including STMicroelectronics and Aristocrat, replacing manual workflows; developed an embedded home-automation prototype for customer demonstrations.

Electronics Corporation of India Limited (ECIL) Dehradun, India Mar 2013 – Jan 2014

Engineer Consultant

- Led a team of 10 operators on the National Population Register project, overseeing daily processing of ~200 sensitive records while enforcing strict data protection and encryption standards.

PROJECTS & VENTURES

Frido · frido.app Side venture Jul 2019 – Jul 2022

Co-owner & Developer

- Spearheaded end-to-end development of a productivity application to minimise cognitive overload, collaborating with UI/UX designers to deliver minimalist interfaces.
- Launched across the Apple ecosystem, achieving a 5-star rating and ranking #2 Product of the Day on Product Hunt.

EDUCATION

MSc Robotics — University of Bristol, UK Sep 2026 – Sep 2027

M.Tech in Software Systems — BITS Pilani, India Jan 2017 – Jan 2019

- Specialisation: Embedded Systems | CGPA: 8.6/10 | First Division
- Dissertation: CAD in Cloud

B.Tech in Electronics & Communication — ICFAI University, Dehradun, India Aug 2008 – May 2012

- CGPA: 9.42/10 | Distinction
- Thesis: Accelerated Parallel Processing Using OpenCL
- Study Project: Wireless Power Transmission

ACHIEVEMENTS & AWARDS

- **Silver Medalist (2nd Rank)** — B.Tech, ICFAI University (2012).
- **Merit Fellowship Recipient (50% Tuition Waiver)** — Awarded for academic excellence across the full duration of B.Tech, ICFAI University (2008–2012).